Sanay Tralshawala

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EDUCATION

Carnegie Mellon University (CMU), Pittsburgh, Pennsylvania **Dec 2026**

Master of Science in Artificial Intelligence Engineering-Mechanical Engineering|Concentration: Robotics & Controls|GPA: 4.00/4.00

Rensselaer Polytechnic Institute (RPI), Troy, New York **May 2025**

Bachelor of Science in Mechanical Engineering | GPA: 3.95/4.00

Dean's Honor List|Rensselaer Leadership Award|Tau Beta Pi Scholarship|Class of 1957 Spectrum Award

EXPERIENCE

Graduate Researcher - Computational Engineering and Robotics Lab (CERLAB), Advisor: Dr. Kenji Shimada **Oct 2025 - Present**

- Project focused on Spatial AI object re-identification for Simultaneous Localization and Mapping (SLAM) system.
- Implementing a Vision Foundation model backbone for latent space embedding and key-point detection..

Visualization and 3D modeling of food while rotating in a Microwave - Computer Vision Engineering **Oct 2025 - Dec 2025**

- Visualize food with thermal imaging to determine the true condition of food while heating to generate cooking suggestions.
- Use of object segmentation methods, including classical frame subtraction approach to mask out color, thermal, and depth data and meshing information together to create a 3D model of food on a rotating microwave plate with temperature data.

Hardware Digital Intern - Northrop Grumman Corporation, Microelectronics Center **Jun 2025 - Aug 2025**

- Designed and developed mechanical assembly of a test fixture using Commercial Off The Shelf parts.
- Performed Structural Analysis on components of test equipment using Solidworks.
- Documented repair and component test procedure, replacement, and troubleshooting process.

RPI Undergraduate Researcher – ISAaC Lab Advisor: Dr. Sandipan Mishra **Jan 2025 - May 2025**

- Symbiotic Human-Autonomy Teaming to model human behavior for a smarter autonomous vehicle guidance system.
- Design, implement, and optimize Model-Predictive Control Algorithm on a Carla simulation.

Prediction of the Power Demand based on Weather Data - Machine Learning Engineering **Jan 2025 - May 2025**

- Analyzed historical weather data in the New York Capital Region and power load on the energy grid.
- Used feature engineering, K-means clustering and hyper-parameter tuning to effectively train and evaluate various models.
- Achieved R-squared of 0.87 for Recurrent Neural Network model and 0.63 for a Random Forest model.

Risk and Reliability Analyst (Student Co-Op) - United States Nuclear Regulatory Commission **Jun 2024 - May 2025**

- Model the plant through PRA and compute the risk of systems and the overall plant stochastically.
- Modify plant models and use SAPHIRE to computationally determine the risk of plant failure due to component failure.
- Assist in developing and validating proposed regulations in technical specifications.

Legislative Intern - Office of Congressman Paul D. Tonko, US House of Representatives **May 2023 - Jun 2023**

- Attended essential briefings and hearings concerning clean energy, electric vehicles, and the shift to sustainable energy
- Contributed to research efforts and crafted informative memorandums addressing high-priority topics, including artificial intelligence, energy optimization, and battery storage.

RPI Undergraduate Researcher – Energy Lab, Advisor: Dr Shankar Narayan **Jan 2023 - May 2023**

- Supported research by conducting experiments to understand flow/heat transfer in a thermodynamic system with complex fluid flow. Assisted in ensuring the validity of automated experiments.
- Recorded experimental data and modified the setup to explore different operating conditions which led to a better understanding of pressure differences due to mass flow rate in a multiphase flow.

LEADERSHIP

Senior Engineering Ambassador (EA) - RPI Engineering Ambassadors **Mar 2023 - May 2025**

- Developed technical presentations for various grade levels and plan & attend school visits to further the mission to inspire future engineers through STEM fundamentals - including modules on power plant/grid efficiency.
- Developed learning modules for the Chip Ambassadors initiative, such as on Chip manufacturing and Chip logic, engaging hands-on activities for semiconductor education, including simulation of the lithography process using nail polish.

Director of Sponsorship - HackRPI Organizing Team **Jan 2023 - Jan 2024**

- Obtained, communicated, and facilitated conversations with 30+ potential sponsors and raised $100,000+ in monetary and prize sponsorship; increased participation from 200 to 500+ with a more diverse attendance for the annual event.

SKILLS

Technical – Computer Vision, Machine Learning, Robot Kinematics, Computer-Aided Design (CAD), Geometric Dimensioning and Tolerancing (GD&T), Probabilistic Risk Assessment (PRA), Numerical Design Optimization, Embodied AI Safety, Agentic AI

Software – LTSpice, Autodesk Inventor & Fusion, Siemens NX, Solidworks, PGAdmin, Docker, Arduino, Nvidia IsaacSim

Programming Languages – Python (TensorFlow, scikit-learn, SciPy, Pandas, OpenCV, Qiskit, PyTorch, PySpark), SQL, R, Java, Matlab/Simulink (Control Systems Toolbox, Optimization Toolbox)

Creative Problem Solving, Critical Thinking, Teamwork, Effective Communication